

AI Action Plan for National Security and Defense

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I. Executive Summary

The future of warfare demands leaner, smarter, and more cost-effective solutions that maximize U.S. military readiness while minimizing waste. Artificial intelligence offers a force-multiplier effect, enabling strategic decision-making at machine speed, reducing operational redundancies, and optimizing force deployment—without excessive increases in personnel, procurement, or logistics. AI-driven simulations and autonomous decision-making tools provide an opportunity to enhance combat effectiveness, reduce costs, and ensure the U.S. maintains a strategic advantage in an era of constrained defense budgets.

The Tech Policy Institute at Cornell is leading this transformation by developing AI agents for wargaming simulations, ensuring that AI-driven battlefield decision-making is stress-tested in dynamic, adversarial environments before live deployment. Unlike traditional defense approaches that rely on costly field exercises and large-scale hardware investments, our high-fidelity simulations, multi-agent AI training, and strategic forecasting allow decision-makers to refine tactics, predict adversary behavior, and adapt to emerging threats at a fraction of the cost.

By integrating technical innovation, doctrinal adaptation, and policy development, the Tech Policy Institute at Cornell ensures that AI-driven warfare is not just cutting-edge—but also cost-effective, scalable, and strategically sound. Our research is designed to help the U.S. military streamline decision-making, optimize force structure, and modernize operations without bloated spending or inefficiencies.

In an era where doing more with less is a national security imperative, AI-driven defense strategies are not just an advantage—they are a necessity. The Tech Policy Institute at Cornell is committed to shaping this future, ensuring that AI innovations drive smarter, faster, and more efficient military capabilities that align with U.S. strategic objectives.

II. The Need for an AI-Driven Military Strategy

AI as a Force Multiplier for the U.S. and Allied Forces

The United States and its allies face increasing operational demands while navigating resource constraints, shrinking force structures, and evolving adversary capabilities. To maintain global military superiority, the U.S. and its partners must move beyond traditional force models and integrate AI as a force multiplier across all domains—land, air, sea, space, and cyber. AI-driven real-time sensor fusion can synthesize intelligence across ISR (Intelligence, Surveillance, and Reconnaissance) platforms, allowing the U.S. and its allies to identify and respond to threats faster than peer competitors. Machine-speed analysis of satellite, drone, and battlefield sensors will give commanders an asymmetric advantage in future conflicts. AI-assisted targeting—integrated with next-generation battle networks—can enhance precision strike capabilities, kinetic and non-kinetic targeting, and force protection, reducing reliance on human reaction time in high-threat environments. Meanwhile, AI-driven predictive logistics and autonomous resupply can increase operational endurance, reducing the vulnerabilities of long supply lines and ensuring rapid force mobilization in contested regions.

For the U.S., AI extends force projection, reduces personnel burden, and accelerates battlefield decision-making, allowing smaller, more agile forces to compete with numerically superior adversaries. For allies, AI integration enhances interoperability and offsets capability gaps, ensuring coalition forces remain resilient in high-intensity operations. These applications are not futuristic; they are necessary today to maintain strategic advantage in a battlespace increasingly defined by speed, adaptability, and data-driven decision-making.

Shifting from Ethical Paralysis to Accelerationism

Current U.S. AI policy is reactive, shaped primarily by ethical concerns, bureaucratic inertia, and adversary-led advancements. While strategic competitors—particularly China—are rapidly deploying AI for both battlefield and hybrid warfare applications, the U.S. has hesitated to operationalize AI beyond narrow, controlled experiments due to fears of automation, transparency concerns, and outdated risk calculations. This risk-averse stance threatens to cede decision dominance to near-peer competitors.

To counter this, the U.S. must abandon the Cold War-era procurement mindset that treats AI as an add-on to existing platforms. AI-enabled warfare fundamentally transforms decision-making, force posture, and operational tempo. The Pentagon must accelerate battlefield AI deployment across strategic, tactical, and autonomous systems rather than waiting for regulatory certainty or adversary escalation. AI is not just another weapons system—it demands entirely new doctrinal and operational models. The focus must shift toward human-machine teaming, where AI enhances warfighter situational awareness, target selection, and adaptive combat decision-making without eliminating human control.

Additionally, AI must be tested not only in simulations and wargames but also in live, high-tempo operational environments. Developing adaptive combat AI systems requires continuous learning and refinement in real-time conflict scenarios. Without this shift, the U.S. risks deploying AI systems that remain untested under real-world pressures, giving adversaries the ability to define the pace and shape of AI-driven warfare. By embracing an accelerationist strategy, the U.S. can dictate the future of AI warfare, ensuring that battlefield AI remains a strategic advantage rather than an existential vulnerability. The question is not whether AI will dominate future conflicts, but who will wield it most effectively—the United States cannot afford to wait.

III. Key AI Policy Actions Needed

Developing Modular, Adaptive AI Architectures

Traditional weapons development follows a platform-centric model, where systems are designed for specific domains (land, air, sea, cyber, space) and updated slowly over time. This approach is incompatible with AI-driven warfare, where adaptability and real-time learning are crucial to maintaining an edge over adversaries. AI systems must be modular, allowing for seamless integration across multiple military domains, and adaptive, capable of continuous battlefield learning and real-time operational adjustments. Furthermore, AI must be interoperable with NATO and allied forces, ensuring that U.S.-led advancements in AI-driven warfare provide a collective advantage rather than an isolated capability.

Institutionalizing Continuous AI Iteration in Military Procurement

The Department of Defense's procurement process remains too slow and rigid to accommodate AI's rapid evolution. The traditional "develop, test, and deploy" model is insufficient when adversaries can update AI-enabled systems in near real-time. To keep pace, the U.S. must transition from static acquisitions to an agile, iterative AI development cycle. This requires establishing real-time battlefield AI feedback loops, allowing AI models to refine their decision-making processes based on live operational data. Continuous software updates must be deployed to already-fielded systems, ensuring they do not become obsolete. Additionally, the U.S. must invest in dynamic testing environments that simulate contested battlespaces, enabling AI to adapt rapidly to evolving threats. AI cannot remain confined to controlled lab environments—it must be tested and evolved in live, high-tempo conflict scenarios.

AI Agents and Wargaming: Stress-Testing Autonomous Warfare

A key element of advancing AI-driven warfare is the development and deployment of AI agents capable of operating in both simulated and real-world military environments. AI agents—autonomous or semi-autonomous systems designed for tactical, operational, and strategic decision-making—serve as critical enablers for real-time decision dominance. These agents can be embedded in command-and-control (C2) systems, battlefield simulations, and autonomous combat platforms,

ensuring the U.S. maintains a competitive edge in both warfighting scenarios and strategic planning exercises.

To accelerate AI battlefield readiness and deployability, the Tech Policy Institute at Cornell proposes leveraging wargaming as a core methodology for refining AI capabilities. Wargaming provides a safe but realistic testing ground for AI-driven decision-making, allowing AI agents to be stress-tested in dynamic, adversarial environments before they are deployed in live operations. By embedding multi-agent AI systems within military simulations, red-teaming exercises, and operational planning scenarios, the U.S. can systematically evaluate AI's effectiveness, adaptability, and resilience against human and machine opponents.

Wargaming also serves as a crucial policy validation tool, enabling military planners and policymakers to anticipate second- and third-order effects of AI warfare strategies. AI-powered wargames allow defense leaders to simulate escalatory scenarios, measure risk in real-time, and refine AI-human coordination protocols to prevent unintended consequences in combat. Additionally, adversarial AI wargaming—where AI systems are pitted against each other in unscripted, evolving conflicts—can expose vulnerabilities and drive continuous iteration in AI warfare algorithms.

For AI to be truly effective in military operations, it must be battle-tested before the battle itself. The Tech Policy Institute at Cornell is uniquely positioned to lead in this domain by integrating AI agents into next-generation military wargaming platforms, providing a critical bridge between technical innovation, strategic doctrine, and operational execution. Through the strategic application of AI agents in combat simulations, the U.S. can ensure that AI-driven warfare is not just an abstract capability but a refined, combat-ready advantage in future conflicts.

Reforming AI Policy Beyond Defensive Diplomacy

The U.S. has historically taken a cautious, reactive approach to military AI, often waiting for global consensus or regulatory frameworks before deploying advanced systems. This defensive diplomacy creates a strategic lag, allowing adversaries—particularly China and Russia—to establish first-mover advantages in AI-driven warfare. Instead of waiting for international AI norms to emerge, the U.S. must set de facto military standards by deploying AI first, shaping how AI is used in warfare through action rather than diplomatic negotiation. AI policy should prioritize operational necessity over abstract ethical debates, recognizing that adversaries are unlikely to abide by the same constraints. Furthermore, the U.S. must strengthen AI coalitions with trusted NATO and Indo-Pacific allies, ensuring that AI-powered military capabilities remain aligned with strategic partners rather than fragmented across national defense priorities.

By adopting a modular, adaptive AI strategy, institutionalizing continuous AI iteration, achieving decision dominance, and shifting from defensive diplomacy to proactive leadership, the U.S. can ensure that AI remains a force multiplier rather than a strategic vulnerability. The battle for AI superiority will not be won through deliberation alone—it will be won through decisive, forward-looking action that defines the future of warfare before adversaries dictate its terms.

IV. The Role of Universities in AI Warfare Development

Universities as AI R&D Incubators

The current AI development landscape is dominated by defense contractors, whose primary focus is on deployment and commercialization rather than foundational research and long-term innovation. While companies like Anduril and Palantir excel at integrating AI into existing military platforms, universities provide a unique interdisciplinary advantage, combining technical research, policy development, and strategic analysis. Cornell, for example, brings together mechanical engineers advancing aerospace AI, political scientists specializing in military strategy and alliances, computer scientists refining AI algorithms for autonomy and decision-making, and business scholars optimizing technology scaling and defense innovation. Unlike industry players operating within commercial constraints, universities can explore next-generation AI capabilities, human-machine teaming models, and doctrinal shifts that will shape the future of AI-driven warfare.

How Universities Can Work with DoD and Industry

To fully leverage this expertise, the U.S. should establish a university-led AI warfighting research consortium in collaboration with the Department of Defense. This initiative would serve as a bridge between academic innovation and military applications, ensuring that cutting-edge AI research translates into operational capabilities. By creating an alternative to contractor-dominated AI development, universities can provide a broader strategic vision—one that aligns emerging AI technologies with evolving military needs rather than just immediate procurement demands. Additionally, universities are well-positioned to ensure that ethical AI discussions remain grounded in operational realities, rather than being dictated by abstract theoretical concerns or regulatory inertia. This approach would allow the U.S. to accelerate AI integration into defense systems while maintaining responsible oversight, ensuring that AI remains both a strategic advantage and a force multiplier for future conflicts.

The Need to Secure AI Research at Universities as a National Security Asset

As artificial intelligence becomes a decisive factor in global power competition, the United States must recognize AI research at universities as a strategic national security asset. However, most top-tier AI research institutions currently lack the infrastructure, governance mechanisms, and security clearances necessary to handle controlled research at scale. This gap leaves critical advancements in AI—particularly in autonomy, cybersecurity, and decision-support systems—vulnerable to foreign exploitation and commercial dilution before they can be fully leveraged for national defense. To address this, the U.S. should establish a formal framework that enables universities to conduct AI research under controlled conditions, ensuring that breakthroughs with military applications remain safeguarded. One approach is to expand the network of University Affiliated Research Centers

(UARCs) specializing in AI, autonomy, and warfare applications. UARCs provide a structured environment where universities can conduct classified or export-controlled research while maintaining academic independence. By designating additional AI-focused UARCs, the U.S. can ensure that universities remain central to AI innovation while enforcing the necessary security protocols to protect sensitive advancements. Without such a mechanism, the U.S. risks ceding control over AI's most transformative military applications, undermining both technological advantage and long-term strategic security.

V. Conclusion: A Call to Action

The future of warfare will be AI-driven, and the United States must dictate its trajectory rather than react to adversary advancements. A bold, accelerationist approach is necessary to ensure that AI is not just a force multiplier but a strategic enabler of U.S. military dominance. While competitors move aggressively to integrate AI into battlefield operations, the U.S. must proactively shape the development, deployment, and governance of military AI rather than waiting for regulatory consensus or external pressures to force action.

A core part of this effort is wargaming-driven AI development—using high-fidelity simulations and multi-agent AI systems to test, refine, and validate AI-driven battlefield decision-making before deployment. AI-powered wargaming provides an adaptive training ground, allowing AI systems to learn from adversarial encounters and adjust tactics dynamically. Without this step, the U.S. risks deploying AI systems that are untested under real-world pressures, giving adversaries the ability to define the pace and shape of AI-driven warfare.

Cornell's Tech Policy Institute is uniquely positioned to lead in this domain, integrating AI research, wargaming technology, and military strategy to develop, deploy, and scale AI-enabled warfighting technologies. To maintain battlefield superiority, the U.S. must not just adopt AI-driven warfare but actively lead its evolution—a mission that demands technological innovation, policy foresight, and rapid battlefield integration.